

Polynomial Functions

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **predict** the end behavior of a polynomial from its degree and leading coefficient.

RELATED SKILLS

I Infer polynomial end behavior from degree and leading coefficient.

A Identify terms, factors, coefficients, variables, and degree in a polynomial expression.

- ☐ 2. I can **determine** polynomial zeros and their multiplicities.

RELATED SKILLS

I Relate zero multiplicity to local graph behavior.

D Determine real and complex zeros using an appropriate method.

- ☐ 3. I can **construct** a polynomial function from specified zeros or graphical features.

RELATED SKILLS

D Build a polynomial from zeros, multiplicities, and a point.

A Determine real and complex zeros using an appropriate method.

A Relate zero multiplicity to local graph behavior.

- ☐ 4. I can **sketch and analyze** a polynomial function using intercepts, multiplicity, and end behavior.

RELATED SKILLS

D Infer polynomial end behavior from degree and leading coefficient.

D Relate zero multiplicity to local graph behavior.

A Determine real and complex zeros using an appropriate method.

- ☐ 5. I can **determine** the minimum possible degree of a polynomial function from the zeros, turning points, inflection points, and end behavior of its graph.

RELATED SKILLS

I Use inflection points to establish a minimum possible polynomial degree.

I Use turning points to establish a minimum possible polynomial degree.

I Use zeros and their multiplicities to establish a minimum possible polynomial degree.

A Infer polynomial end behavior from degree and leading coefficient.

☐ **6. I can **build or select** a polynomial model for a context and interpret its features.**

RELATED SKILLS

- ☐ D Explain a model parameter in the language of its context.
 - ☐ D Select a function family whose behavior fits a table, graph, or context.
 - ☐ A Build a polynomial from zeros, multiplicities, and a point.
 - ☐ A Restrict a model to meaningful contextual inputs.
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